

require 802.1q trunk.

- Full support for the SCTP (stream control transmission protocol).
- Provides high availability to the SDN network via the allowed address pair feature.
- Port health checking.

Official website: <http://www.opencontrail.org/>

Latest version: 4.0.1

Open Network Operating System (ONOS)

ONOS is an open source SDN platform for service providers, designed and developed by ON.Lab. The primary goal is to create an SDN operating system that enables communications service providers to provide scalability, efficient performance and high availability.

It is designed for building next-generation SDN/NFV solutions, and provides the flexibility to create and deploy new dynamic network services with a simple programming interface. It supports both configuration and real-time control of the network, eliminating the need to run routing and switching control protocols inside the network fabric.

The platform provides applications with a number of high-level abstractions, through which the applications can learn about the state of the network and through which it can control the flow of traffic through the network. The network graph abstraction provides information about the structure and topology of the network. The flow objective is a device-centric abstraction that allows applications to direct the flow of traffic through a specific device without the need to be aware of the device table pipeline.

Features

- ONOS can be deployed as a service on a cluster of services, and ONOS software can run on each server. It makes ONOS highly scalable and facilitates seamless capacity.
- ONOS features northbound and southbound application program interfaces (APIs) grounded in abstraction to prevent configuration and protocol lock-in for applications and devices, respectively. ONOS uses its Intent Framework subsystem that allows applications to specify what they need from the system—if an application needs more bandwidth.
- ONOS Core is distributed to provide reachability to each network switching device. The ONOS controller remains logically centralised and the separate subdivisions or instances in the complete ONOS architecture can be viewed and accessed as a single system.
- Software modularity in ONOS means that the community has been diligent about keeping software functions well defined and localised by defining the right abstractions and interfaces. This has many

important benefits, including software that is easier to read, test and maintain.

- ONOS abstracts device characteristics so that the core operating system does not have to be aware of the particular protocol being used to control or configure a device. ONOS has an extensive and growing list of southbound support including P4, OpenFlow, NETCONF, TL1, SNMP, CLI, BGP, RESTCONF and more.
- The ONOS YANG tool-chain provides a compiler capable of parsing YANG source files and generating Java artefacts, which can be used for writing applications against the abstractions defined by the YANG models. It also provides a runtime capable of encoding and decoding between such internal models and their external JSON or XML data representations.
- Other features include high performance, resilience, next-gen device support and legacy device support.

Official website: <https://onosproject.org>

Open vSwitch

Open vSwitch is an open source implementation of the distributed virtual multi-layer switch, licensed under the open source Apache 2 licence. It was originally created by a team at Nicira and now it has been acquired by VMware. Open vSwitch is designed to enable effective network automation through programmatic extensions, while supporting standard management interfaces and protocols like NetFlow, sFlow, SPAN, RSPAN, CLI, LACP and 802.1ag.

Tools provided by Open vSwitch

- **ovs-ofctl:** For querying and controlling OpenFlow switches and controllers.
- **ovs-pki:** For creating and managing public-key infrastructure for OpenFlow switches.
- **ovs-testcontroller:** This is useful for testing.

Features

- **IPv6 support:** Multiple tunnelling protocols - GRE, VXLAN, STT with IPsec support.
- **Best QoS:** Support for HFSC qdisc; kernel and user-space forwarding engine options.
- **BFD and 802.1ag link monitoring:** Standard 802.1Q VLAN model with trunking.
- Transactional configuration database with C and Python bindings.
- Forwarding layer abstraction to ease pointing to new software and hardware platforms.

Official website: <http://www.openvswitch.org>

Latest version: 2.9.0